



AADIT AGRAWAL

Codex Consultant

AI/ML · HPC · DevOps · Security

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LANGUAGES
Python, C/C++, TypeScript,
JavaScript, Rust, Go, Zig,
Bash/Zsh, SQL, Java, Typst

FRONTEND / BACKEND
Next.js, React, Svelte,
Astro, Tailwind, Radix ·
Node.js, Bun, tRPC, FastAPI

DEVOPS / NETWORKING
Docker, Proxmox, Linux,
GitHub Actions, Ansible,
Terraform, Tailscale, Nginx,
Caddy, Grafana, Prometheus,
Loki

AI / ML
PyTorch, JAX, Transformers,
OpenCV, YOLO, SGLang, vLLM,
llama.cpp, TensorRT-LLM, MLX,
CUDA, RDMA, NCCL

AGENTIC TOOLING
Claude Code, Codex, Pi.dev,
Droid, Cursor, opencode, MCP,
CodeRabbit

DATABASES / SECURITY
PostgreSQL, SQLite, Redis,
pgvector · Wireshark, Burp,
Nmap, Ghidra

EDUCATION
Manipal Institute of
Technology

BTech, Computer and
Communication Engineering
Jul 2023 - May 2027

EDUCATION

Manipal Institute of Technology · BTech - CCE, MAHE

Jul 2023 - May 2027

EXPERIENCE

Codex Consultant · OpenAI India

May 2026 - Present

- Consulting engagement focused on Codex and agentic developer workflows.

Head Student Researcher · CASIR, MAHE - ecitis.org

May 2025 - Present

- Architected on-prem inference/HPC stack serving 1,500+ daily requests at 99.2% uptime on SGLang/vLLM/llama.cpp.
- Configured a first-of-its-kind-in-India 4x Mac Studio cluster (512GB unified memory, 16TB storage) with Thunderbolt RDMA / InfiniBand-style fabric for MLX/MPS inference and training; leading USD 300K NVIDIA Blackwell bring-up.
- Benchmarking PyTorch, JAX, vLLM, SGLang, TensorRT-LLM, CUDA, NCCL, and MLX runtimes for heterogeneous on-prem inference and training workloads.
- Delivered production AI systems through CASIR's Schneider Electric collaboration: PLC code translation (70% migration-effort reduction), retrieval over 2,000+ docs (94% accuracy, 3x faster), and app-security analysis (3 weeks to 3 days).
- Built real-time CV on NVIDIA Jetson + DJI drone at 30fps edge inference; 2 peer-reviewed publications in progress.

Web and Design Engineer · [Rossmann New Media 2 Inc.](https://rossmannnewmedia2.com)

Jan 2026 - Present

- Freelance/contract role redesigning and redeveloping rossmannnewmedia2.com: WordPress to Next.js rebuild, responsive UI, performance fixes, and production deployment support.
- Technical SEO work for data-recovery queries moved primary rankings from 100+ to the top 5; earlier 2022-2024 work included thumbnails, posts, and merchandise for million+ view Rossmann media.

DevOps/MLOps Engineering Intern · Pixa AI, Bengaluru

Dec 2025

- Provisioned H100/H200/B200 GPU compute and CI/CD pipelines for Phi-4 TTS/STT fine-tuning; supported 2 Model API releases at 99.5% deployment success.
- Debugged distributed training and deployment issues across NCCL, drivers, containers, and runtime configuration.
- Improved release handoff between model experiments and API deployment by standardising environment setup, GPU allocation, and failure triage notes.

RESEARCH / WRITING

FL + Digital Twin Route Optimisation · SCE, MAHE

Jun 2024 - Jul 2025

- Built GPS telemetry pipeline over 500K+ datapoints and dashcam CV dataset with OpenCV + YOLO over 12,000+ labelled samples.
- Implemented privacy-preserving federated-learning pipeline for distributed route optimisation; evaluated client drift and non-IID convergence behaviour.

ECITIS Technical Blog · blog.ecitis.org

2025 - Present

- Authored 18 long-form guides on TensorRT/Triton serving, Apple MLX/Metal, ROCm, RLHF/DPO, MoE, edge AI, VLMs, coding agents, and HPC cluster tuning.

PROJECTS

Homelab Infrastructure · Self-hosted cloud

2022 - Present

- Operate 70+ services on Proxmox/Docker with 98% uptime, 8TB MergerFS, Grafana/Prometheus/Loki monitoring, 3-2-1 backups, self-hosted git/blog/design services, and GPU media/AI workloads.
- Fine-tuned/deployed object-detection models on security-camera feeds at 30fps and offloaded 95% of 4K media/AI compute from CPU to GPU.
- Mitigated live DDoS traffic at 50K+ req/min using fail2ban, geo-blocking, segmentation, mTLS, and auth proxies; reduced attack surface by 95%.

The TIK Menu · Campus food platform

2025 - Present

- Built zero-cost campus menu platform with 5,000+ DAU, 150K+ monthly requests, dynamic XLSX parsing, Cloudflare R2 storage, REST API, PWA/offline support, and campus-wide adoption in under 7 days.

Coolstuff.work + OSS Tooling · git.aadit.cc

2023 - Present

- Built utility suite reaching 10,000+ aggregate DAU: OR-Free model/pricing tracker, MIT Courses Search (500+ graph nodes), Campus Directory, Contact Cards, Mass Contact Maker, and transcript-cli.
- Maintained mostly zero-cost deployments across Cloudflare, self-hosted Git, and static/PWA surfaces with fast iteration loops for campus-scale tools.

Agent + Developer Tooling · OSS / self-hosted

2024 - Present

- Published agentskills, a t3code fork, and ophoner mobile agent loop; built tmux + Android shell workflows for parallelised coding-agent experiments.
- Built Typsmthng, a Typst editor with WASM preview and vim-mode experiments; run self-hosted CI/sandboxing and CodeRabbit-backed review flows for public repos.

LEADERSHIP

Founding President · [PRISM](https://prism.mahe.ac.in)

Jan 2026 - Present

- Founded 150+ member interdisciplinary student research club; serve on MAHE Academic Advisory Board for AI Curriculum and Adoption.
- Coordinate research groups across AI, systems, product, and design; maintain PRISM's web presence, GitHub organisation, and student-facing technical material.

Student Society Leadership · MIST, Hult Prize, Astronomy Club, TechTatva

2023 - 2025

- Led student security, design, and event work: CTF infra for 300+ users at 100% uptime, 12+ workshops for 150+ participants, Hult Prize graphics, Astronomy Club outreach, and TechTatva CAP organizing.
- Built club web properties, security-awareness material, and event communications; competed in national CTFs with top-10% finishes.